

Operating System Exercises

1. The two main functions of an operating system are:

- Resource management: Managing computer hardware resources including CPU, memory, disk drives, and peripherals
- Providing a user interface: Creating an environment for users to interact with the computer system, either through a graphical user interface (GUI) or command-line interface (CLI)

2. Applications for nine different types of operating systems:

- Batch operating systems: Payroll processing systems
- Time-sharing operating systems: University computer labs
- Distributed operating systems: Telecommunications networks
- Network operating systems: Corporate file sharing environments
- Real-time operating systems: Air traffic control systems
- Mobile operating systems: Smartphones and tablets
- Multiprocessor operating systems: Scientific computing centers
- Multiprogramming operating systems: Data processing centers
- Embedded operating systems: Smart home appliances

3. The difference between timesharing and multiprogramming systems:

- Multiprogramming systems focus on maximizing CPU utilization by keeping multiple programs in memory simultaneously, allowing the CPU to switch between them when one is waiting for I/O operations.
- Timesharing systems focus on minimizing response time and providing interactive capabilities to multiple users simultaneously. They allocate fixed time slices to different processes and switch between them frequently, creating the illusion that each user has dedicated access to the system.